

GLASGOW PRESTWICK RNAS, United Kingdom

WMO: 031360

Lat: 55.5153N

Lon: 4.5847W

Elev: 27

StdP: 101.00

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.2	(d) -6.2	(e) 2.2	(f) -1.3	(g) -4.8	(h) 2.5	(i) -0.3	(j) 18.0	(k) 7.6	(l) 15.7	(m) 7.9	(n) 2.0	(o) 80	(p) 0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.4	(c) 22.7	(d) 17.2	(e) 20.8	(f) 16.2	(g) 19.3	(h) 15.5	(i) 18.0	(j) 21.4	(k) 17.0	(l) 19.7	(m) 16.2	(n) 18.5	(o) 4.3	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.7	(b) 11.9	(c) 19.4	(d) 15.8	(e) 11.3	(f) 18.3	(g) 15.0	(h) 10.7	(i) 17.4	(j) 50.9	(k) 21.3	(l) 47.9	(m) 19.8	(n) 45.4	(o) 18.6	(p) 26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	12.0	10.5	-6.0	26.3	2.0	2.2	-7.4	27.9	-8.5	29.3	-9.6	30.5	-11.1	32.2
			-6.3	20.2	2.0	1.8	-7.8	21.5	-8.9	22.5	-10.1	23.6	-11.5	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	4.9	5.1	6.2	8.2	10.8	13.4	15.1	15.0	13.2	10.4	7.3	5.2
	DBStd	4.61	2.95	2.77	2.58	2.58	2.69	2.19	2.05	2.02	2.19	2.65	3.02	3.66
	HDD10.0	769	158	138	119	64	20	1	0	0	2	27	87	152
	HDD18.3	3192	416	371	377	303	232	149	103	105	154	245	330	408
	CDD10.0	626	1	1	2	11	47	103	158	157	98	41	7	2
	CDD18.3	7	0	0	0	0	0	1	3	3	1	0	0	0
	CDH23.3	46	0	0	0	0	4	8	21	12	1	0	0	0
	CDH26.7	5	0	0	0	0	0	1	3	1	0	0	0	0
(14) Wind	WSAvg	4.8	5.5	5.5	5.1	4.6	4.5	4.2	4.0	4.1	4.5	4.9	5.0	5.3
(15) Precipitation	PrecAvg	949	91	65	73	51	54	61	69	85	103	110	105	96
	PrecMax	1210	183	180	135	131	122	142	164	172	245	203	182	194
	PrecMin	705	12	4	18	4	7	21	14	18	21	35	26	22
	PrecStd	125	43	39	35	25	26	23	27	43	50	50	43	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.7	14.3	19.1	23.1	24.3	26.2	25.2	21.2	17.5	14.7	13.2
		MCWB	10.7	10.8	10.4	12.5	15.6	17.0	18.8	19.3	16.4	14.7	13.1	11.8
	2%	DB	10.8	11.1	12.0	15.5	19.8	21.0	22.6	21.7	19.2	16.0	13.4	12.1
		MCWB	9.6	9.4	9.4	10.9	14.3	15.7	17.4	17.4	15.6	13.8	12.0	10.7
	5%	DB	9.9	10.1	10.9	13.6	17.1	19.0	20.5	19.8	17.7	15.1	12.4	11.0
		MCWB	8.8	8.7	8.8	10.3	12.7	15.0	16.4	16.4	14.9	13.2	11.0	9.8
	10%	DB	9.2	9.2	10.0	12.3	15.3	17.4	18.8	18.4	16.6	14.2	11.6	10.0
		MCWB	8.0	7.8	8.2	9.4	11.8	13.9	15.5	15.5	14.1	12.4	10.2	8.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.0	11.2	13.2	16.8	18.0	19.6	20.0	17.6	15.4	13.5	12.0
		MCDB	11.9	12.3	13.1	17.3	21.4	22.1	24.2	23.9	20.2	16.8	14.4	12.9
	2%	WB	9.9	9.8	10.2	11.9	14.8	16.6	18.0	18.0	16.4	14.3	12.2	10.9
		MCDB	10.7	10.7	11.5	14.7	18.5	20.0	21.6	20.9	18.3	15.5	13.2	11.9
	5%	WB	9.0	8.9	9.2	10.7	13.4	15.3	16.9	16.9	15.4	13.5	11.3	9.9
		MCDB	9.8	9.9	10.5	13.0	16.1	18.1	19.7	19.2	17.1	14.7	12.2	10.9
	10%	WB	8.1	8.0	8.5	9.8	12.3	14.4	16.0	16.1	14.6	12.6	10.4	8.9
		MCDB	9.1	9.1	9.7	11.9	14.8	16.8	18.3	17.9	16.2	14.0	11.4	10.0
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	5.8	6.8	7.5	6.7	6.4	6.4	6.0	5.6	5.1	5.0
		MCDBR	4.4	5.1	6.5	8.9	11.0	9.4	9.4	8.4	7.4	5.7	4.8	4.9
	5% WB	MCWBR	4.4	4.4	4.7	5.6	6.4	5.5	5.4	5.0	4.6	4.4	4.5	4.8
		MCWBR	4.5	4.7	5.6	7.7	9.5	8.4	8.0	7.5	6.3	5.1	4.7	5.1
(40) Clear-Sky Solar Irradiance	taub		0.338	0.347	0.372	0.390	0.393	0.387	0.388	0.373	0.365	0.347	0.327	0.323
		taud	2.202	2.298	2.272	2.261	2.303	2.353	2.357	2.412	2.430	2.404	2.290	2.183
	Ebn at Noon		550	713	786	829	849	858	848	840	794	710	568	470
		Edh at Noon	61	79	101	117	119	115	113	100	86	70	55	49
(44) All-Sky Solar Radiation	RadAvg		0.49	1.18	2.31	3.73	4.78	4.70	4.50	3.65	2.54	1.37	0.66	0.37
	RadStd		0.07	0.13	0.17	0.29	0.44	0.51	0.45	0.27	0.20	0.19	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page